Parameter Estimation

Us.

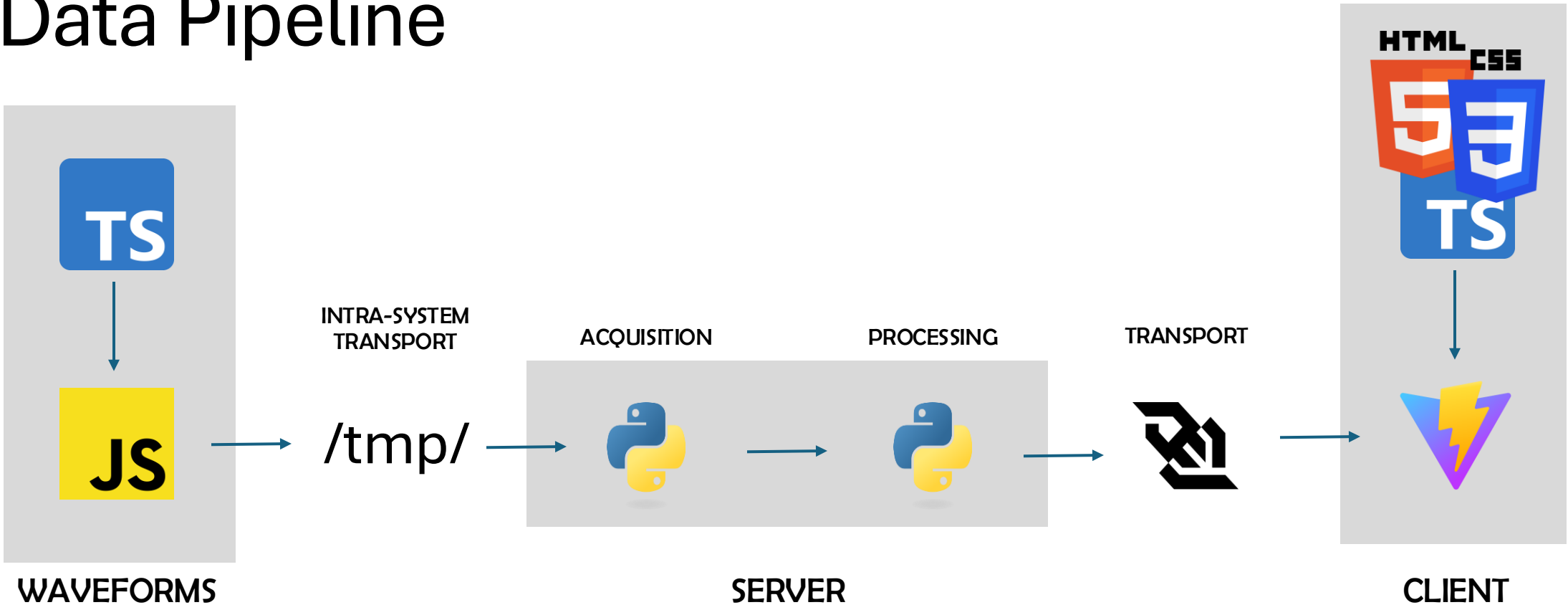
Goal

- Improve the existing signal-processing algorithm using an FFT
- Remove the need for a known input frequency
- Automatically estimate and continuously update the signal frequency
- Calculate amplitude, phase, and offset from the measured signal
- Use mean, standard deviation, and error to compare each algorithm

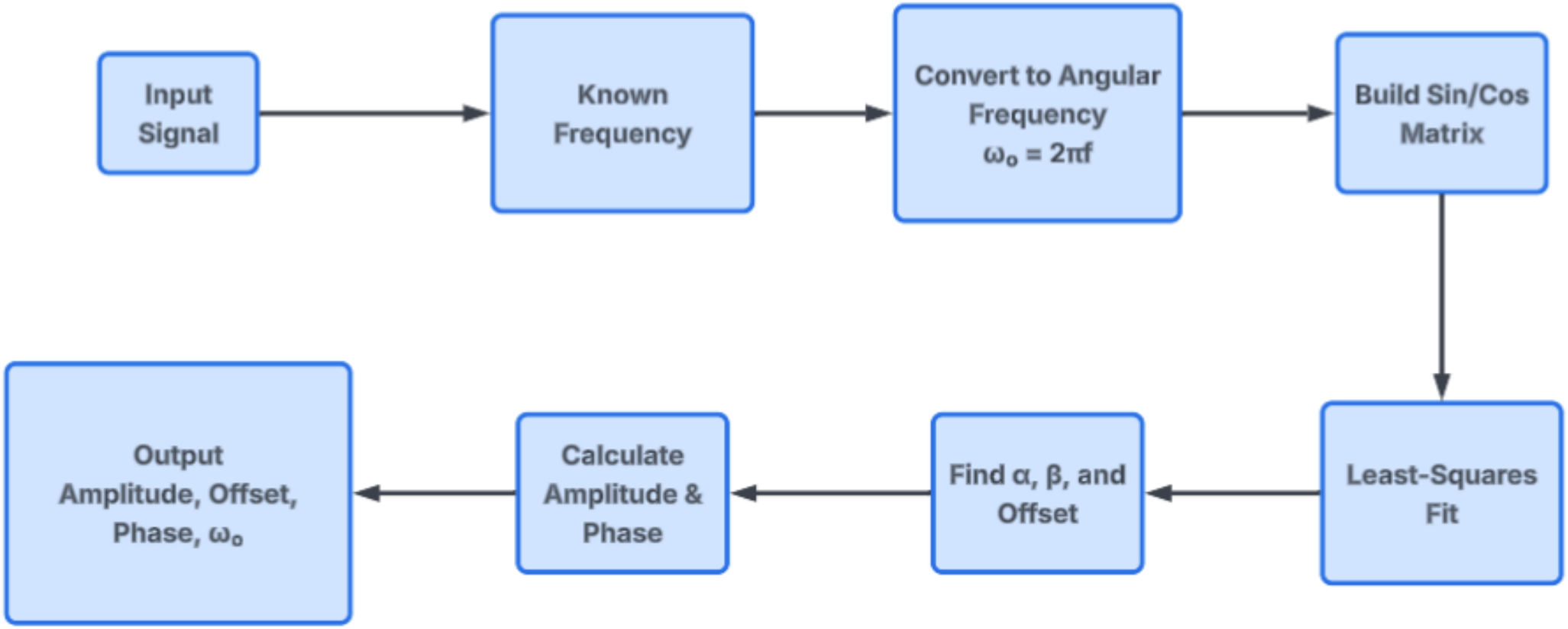
Hardware

- The AD2 was used to generate and measure the test waveform
- An oscilloscope was also connected to the AD2 to inject an external noisy signal at different frequencies

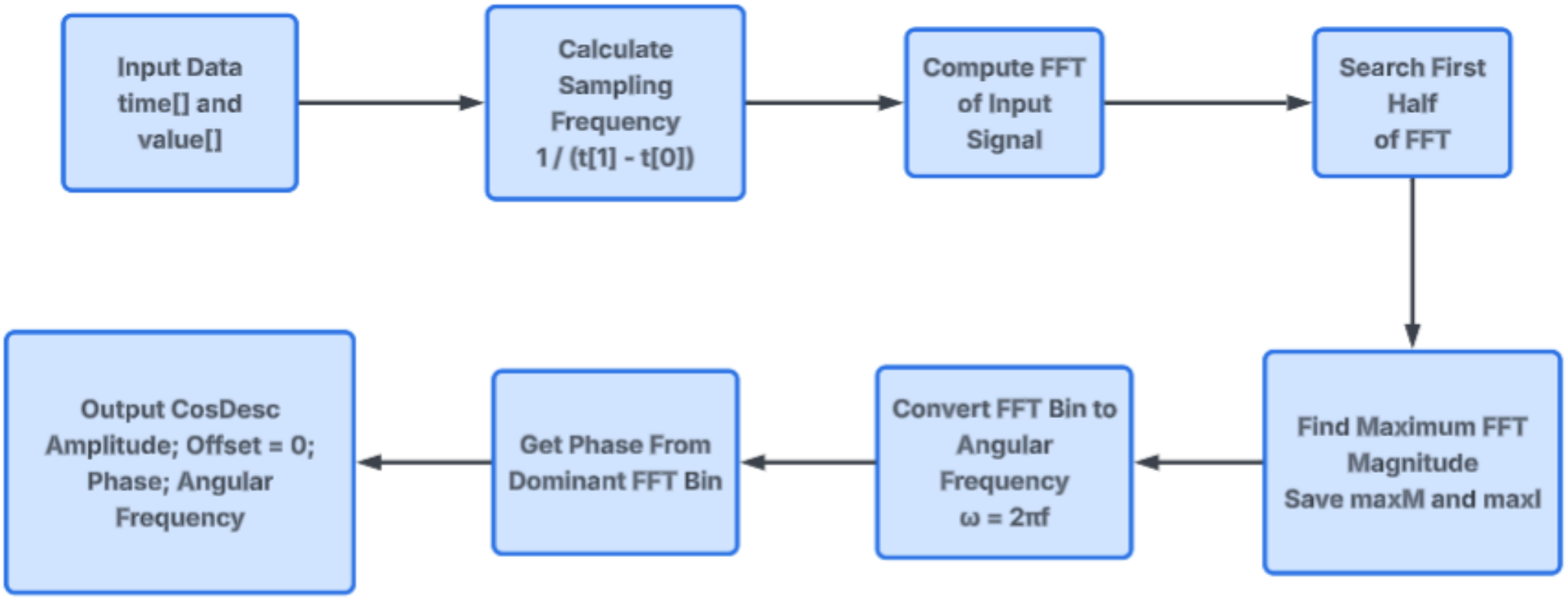
Data Pipeline



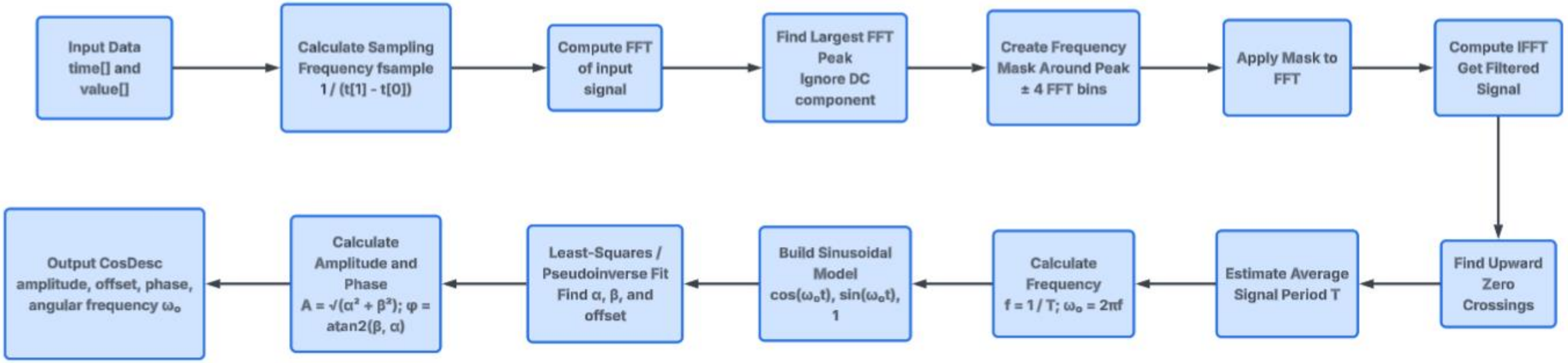
Original Algorithm



Iteration 1

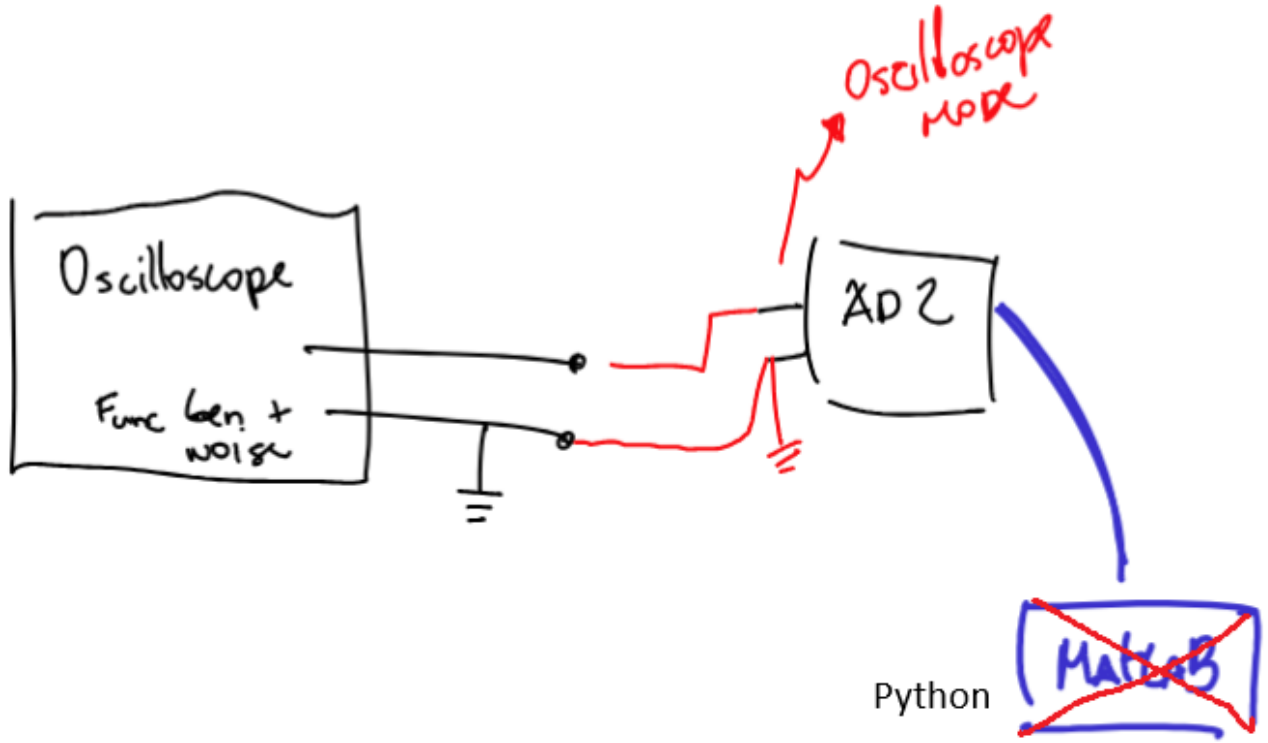

Input Data time[] and value[]
Calculate Sampling Frequency 1 / (t[1] - t[0])
Compute FFT of Input Signal
Search First Half of FFT
Find Maximum FFT Magnitude Save maxM and maxI
Convert FFT Bin to Angular Frequency ω = 2πf
Get Phase From Dominant FFT Bin
Output CosDesc Amplitude; Offset = 0; Phase; Angular Frequency

Final Iteration



Methodology

- Tested each algorithm across different frequencies
- Added varying noise to waveforms
- Repeated the same tests using the oscilloscope connected to the AD2
- Compare the results of each algorithm using the mean, standard deviation, and error



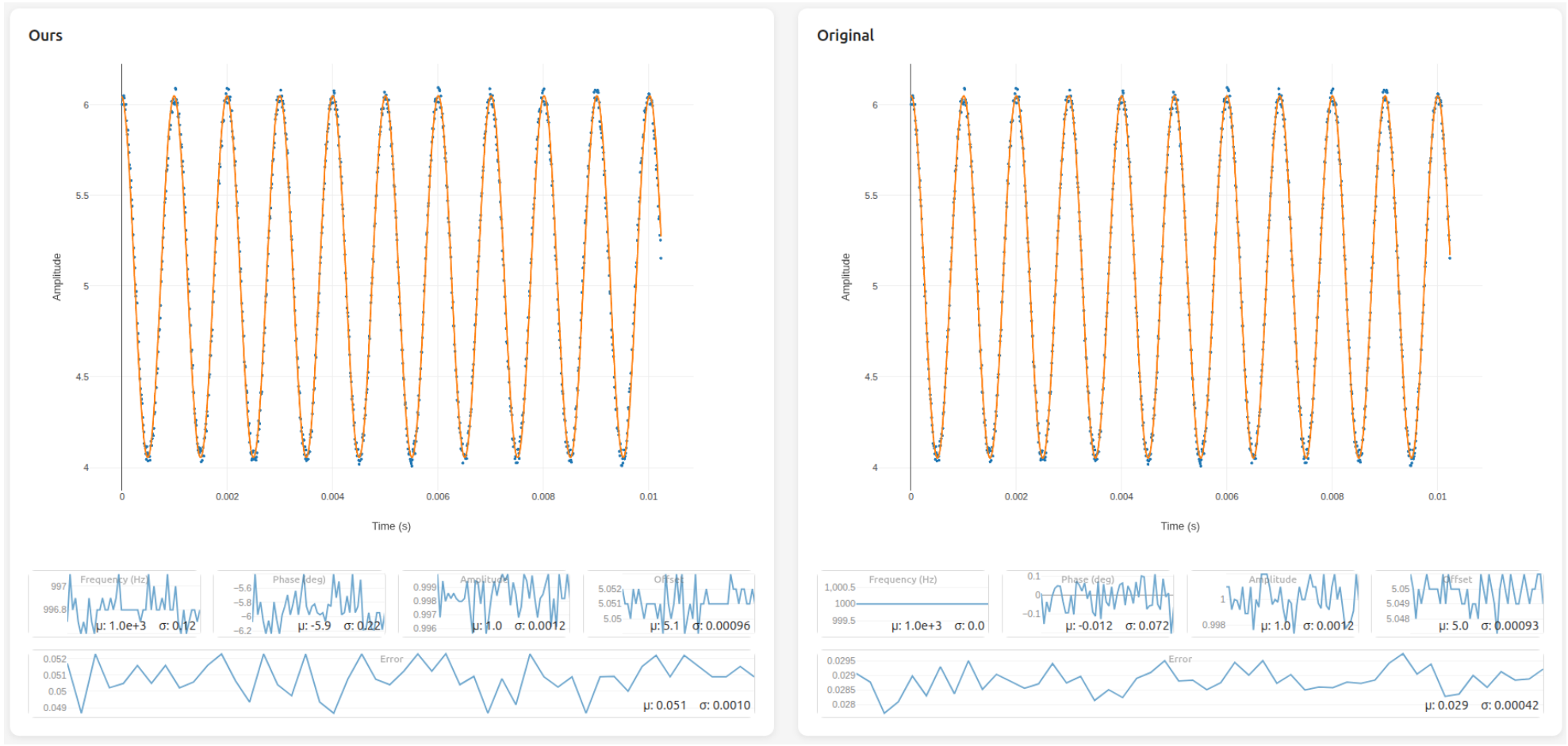
Live Demo



10.4.53.37:5173/

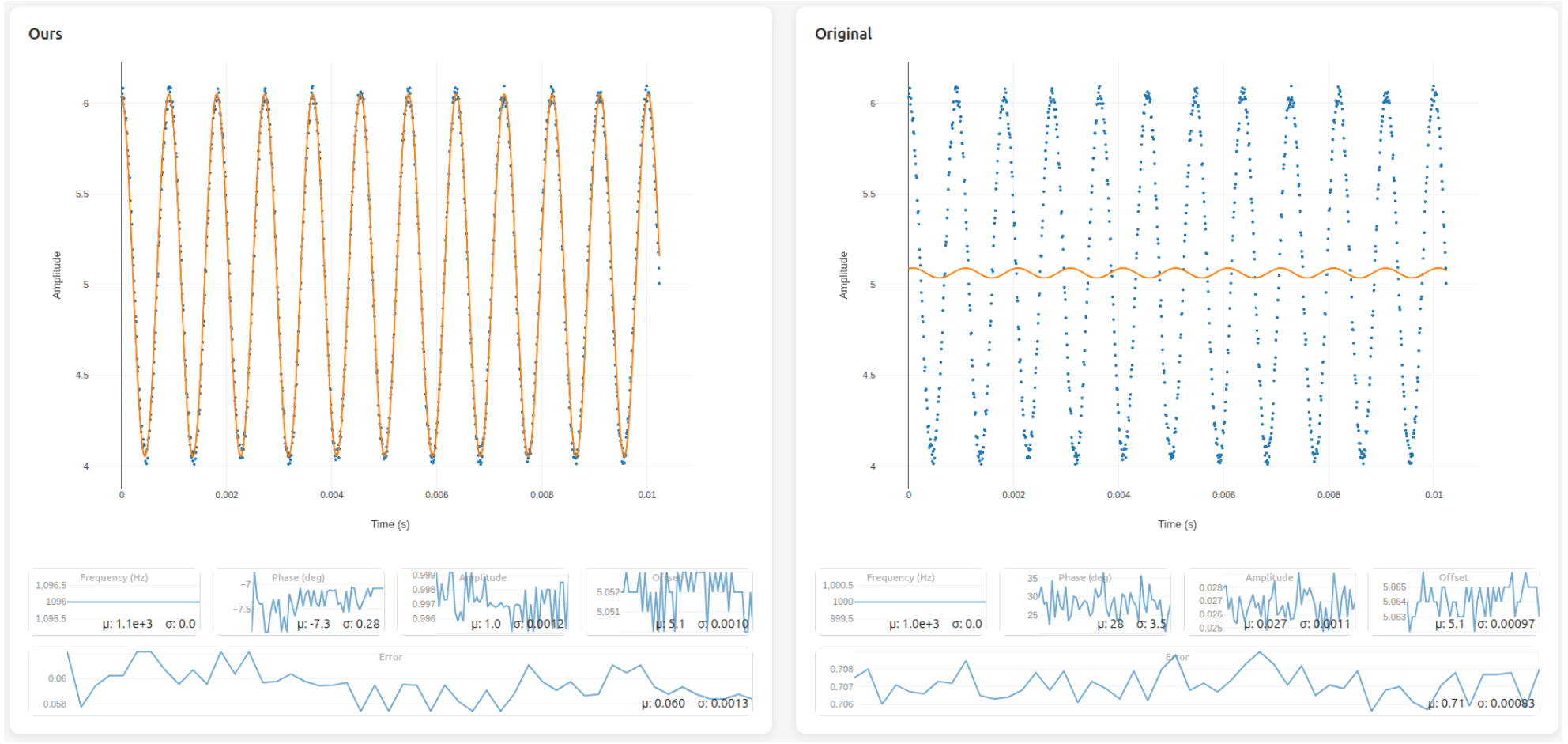
NOT A VIRUS

Results



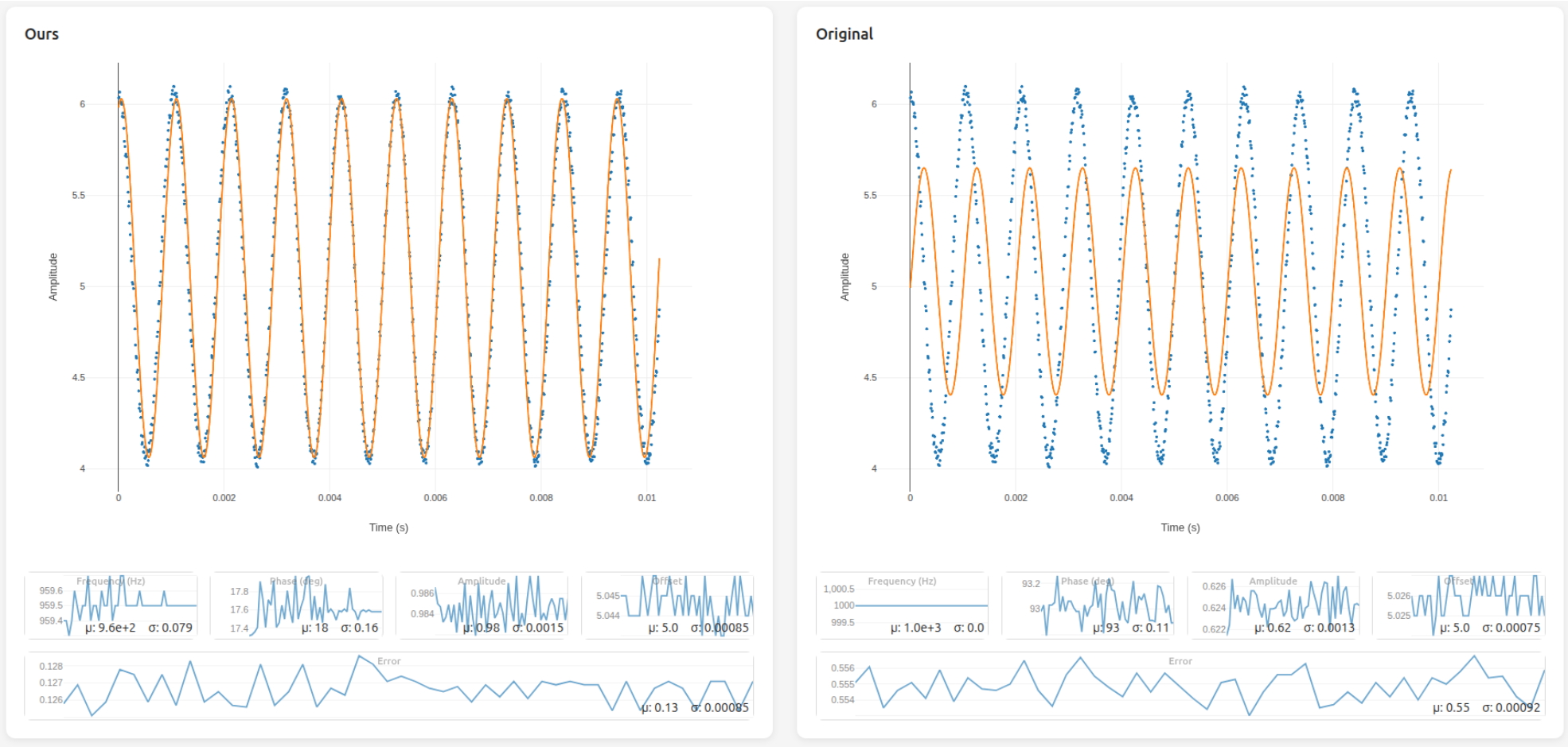
Frequency: 1kHz
Noise: 10%

Results



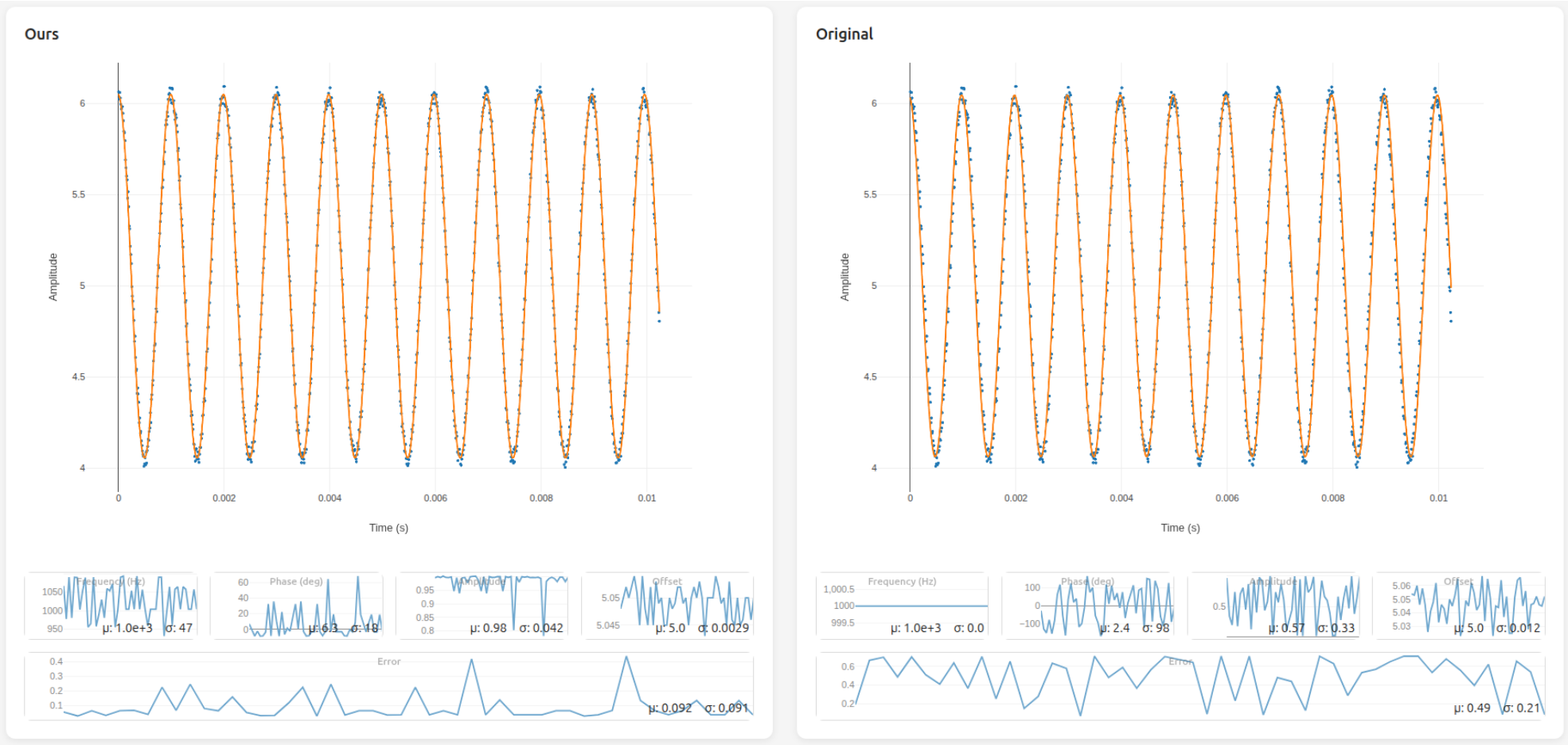
Frequency: 1.1kHz
Noise: 10%

Results



Frequency: 900Hz
Noise: 10%

Results



Frequency: 0.8kHz - 1.2kHz
Noise: 10%

Graphic Design is my Passion

